

① Kernel Transformation.

given kernel function,

$$K(x, x') = (1 + x^T x')^2$$

As we know this kernel function lies under the example of polynomial.

So, Polynomial of power P : $k(x, x') = (1 + x^T \cdot x')^P$

for mapping $\phi: x \rightarrow \phi(x)$, where $\phi(x)$ has $\left(\frac{d+P}{P}\right)$ dimensions.

The function $\phi: x \rightarrow z$ is a polynomial of order P

Kernel function: $k(x_i, x_j) = (1 + x_i^T \cdot x_j)^P$.

$$\text{Dimension of } z = \left(\frac{d+P}{P}\right)$$

$$d = 3 \Rightarrow P = 2.$$

$$z = \left(\frac{3+2}{2}\right) = \left\lceil \frac{5}{2} \right\rceil = \frac{5 \cdot 4}{2 \cdot 1} = 10$$

$$z = 10.$$

$$\phi(x) = [1, x_1^2, \sqrt{2}x_1, x_2, x_2^2, \sqrt{2}x_2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_2]^T$$

$$\Rightarrow [1, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_3, x_1^2, x_2^2, x_3^2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_2, \sqrt{2}x_1 x_2]$$

$$\therefore \phi(x) = [1, x_1^2, x_2^2, x_3^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_3, \sqrt{2}x_1x_2, \sqrt{2}x_1x_3, \sqrt{2}x_2x_3]$$

②. For knowing the probability that a sequence HHTT is generated by the model when the hidden path is FBBF.

We have to calculate HHTT sequence through the hidden path sequence which is present in given in the question.

From given sequence,

$$P(\text{HHTT} | \text{FBBF}) = P(H|F) \cdot P(H|B)$$

$$P(T|B) \cdot P(T|F)$$

$$= 0.5 \times 0.5 \times 0.4 \times 0.75 \times 0.6 \times 0.25 \times 0.4 \times 0.5$$

$$= 0.25 \times 0.3 \times 0.15 \times 0.2$$

$$= 0.00225$$

So, the final probability of sequence HHTT hidden path with FBBF can be viewed as 0.00225.

③. Part A: Mutual Information (MI)

$$\text{Formulae: } I(A, B) = \sum_{a, b} P(a, b) \cdot \log \frac{P(a, b)}{P(a) \cdot P(b)}$$

for probabilities income, gender, age & risk.

$$P(\text{Gender M}) = 4/8 = 1/2$$

$$P(\text{Gender F}) = 4/8 = 1/2$$

$$P(\text{income} > 50) = 5/8$$

$$P(\text{income} < 50) = 3/8$$

$$P(\text{risk, high}) = 5/8$$

$$P(\text{risk, low}) = 3/8$$

$$P(\text{Age} \leq 25) = 4/8 = 1/2$$

$$P(\text{Age} > 25) = 4/8 = 1/2$$

→ Mutual information for age

$$P(\text{Age} > 25, \text{high}) = 1/8$$

$$P(\text{Age} > 25, \text{low}) = 3/8$$

$$P(\text{Age} \leq 25, \text{high}) = 1/2$$

$$P(\text{Age} \leq 25, \text{low}) = 0$$

$$P(\text{Age} > 25, \text{high})$$

$$= 1/8 \log(1/8 / (1/2 \times 5/8))$$

$$= 1/8 \log(0.125 / (0.5 \times 0.625))$$

$$= 1/8 \log(0.398)$$

$$= -0.0500$$

$$\begin{aligned}
 P(\text{Age} > 25, \text{low}) &= 3/8 \log \left(\frac{3/2}{1/2} \times 3/8 \right) \\
 &= 3/8 \log (0.375 / 0.875) \\
 &= 0.1128
 \end{aligned}$$

$$\begin{aligned}
 P(\text{Age} \leq 25, \text{high}) &= 1/2 \log \left(\frac{1/2}{1/2} \times 5/8 \right) \\
 &= 1/2 \log (0.5 / 0.5 \times 0.625) \\
 &= 1/2 \log (0.5 / 0.3125) \\
 &= 0.5 \times 0.2041 \\
 &= 0.10205
 \end{aligned}$$

0.1648 is the mutual information of age

→ For income:

$$P(\text{income} > 50, \text{high}) = 2/8 = 1/4$$

$$P(\text{income} > 50, \text{low}) = 3/8$$

$$P(\text{income} < 50, \text{high}) = 3/8$$

$$P(\text{income} < 50, \text{low}) = 0/8$$

$$\begin{aligned}
 P(\text{income} > 50, \text{high}) &= 2/8 \log \left(\frac{1/4}{(5/8) \times (5/8)} \right) \\
 &= 1/4 \log (0.25 / 0.390) \\
 &= -0.0482
 \end{aligned}$$

$$\begin{aligned}
 P(\text{income} > 50, \text{low}) &= 3/8 \log \left(\frac{3/8}{5/8 \times 3/8} \right) \\
 &= 3/8 \log (0.375 / 0.625 \times 0.375) \\
 &= 0.0765
 \end{aligned}$$

$$\begin{aligned}
 P(\text{income} < 50, \text{high}) &= \frac{3}{8} \log \left(\frac{3}{8} \middle| \frac{3}{8} \times \frac{5}{8} \right) \\
 &= \frac{3}{8} \log (0.375 \mid 0.625 \times 0.375) \\
 &= 0.765
 \end{aligned}$$

$$P(\text{income} < 50, \text{low}) = 0 \times \log \left(0 \middle| \frac{3}{8} \times 0 \right) = 0$$

final mutual information of income is 0.1047.

→ for gender:

$$P(M, \text{high}) = \frac{1}{4}$$

$$P(M, \text{low}) = \frac{1}{4}$$

$$P(F, \text{high}) = \frac{3}{8}$$

$$P(F, \text{low}) = \frac{1}{8}$$

then

$$P(M, \text{high}) = \frac{1}{4} \log \left(\frac{1}{4} \middle| \frac{1}{2} \times \frac{5}{8} \right)$$

$$= \frac{1}{4} \log (0.25 \mid 0.5 \times 0.625)$$

$$= -0.0242$$

$$P(M, \text{low}) = \frac{1}{4} \log \left(\frac{1}{4} \middle| \frac{1}{2} \times \frac{3}{8} \right)$$

$$= \frac{1}{4} \log (0.25 \mid 0.5 \times 0.375)$$

$$= 0.0312$$

$$P(F, \text{high}) = \frac{3}{8} \log \left(\frac{3}{8} \middle| \frac{1}{2} \times \frac{5}{8} \right)$$

$$= \frac{3}{8} \log (0.375 \mid 0.5 \times 0.625)$$

$$= 0.0296$$

$$P(F, \text{low}) = 1/8 \log \left(\frac{1/8}{1/2 \times 3/4} \right) \\ = 1/8 \log (0.125 / 0.15 \times 0.375) \\ = -0.02255$$

Mutual information for gender is 0.0141

Part B : Chi - Square Test.

$$\text{Formulae : } \chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

→ For age, the observed frequencies are:

$$O(\text{Age} > 25, \text{high}) = 1$$

$$O(\text{Age} > 25, \text{low}) = 3$$

$$O(\text{Age} \leq 25, \text{high}) = 4$$

$$O(\text{Age} \leq 25, \text{low}) = 0$$

expected frequency:

$$E = 4 \times 5 / 8 = 2.5$$

$$E = 4 \times 3 / 8 = 1.5$$

$$E = 4 \times 5 / 8 = 2.5$$

$$E = 4 \times 3 / 8 = 1.5$$

$$\chi^2 = (1 - 2.5)^2 / 2.5 + (3 - 1.5)^2 / 1.5 + (4 - 2.5)^2 / 2.5 + (0 - 1.5)^2 / 1.5 \\ = 4.8$$

$$\text{Chi-Square} = 4.8$$

→ for income, observed frequencies are:

$$O(\text{income} > 50, \text{high}) = 2$$

$$O(\text{income} > 50, \text{low}) = 2$$

$$O(\text{income} < 50, \text{high}) = 3$$

$$O(\text{income} < 50, \text{low}) = 0$$

Expected frequencies:

$$E = 5 \times 5 / 6 = 3.125$$

$$E = 15 / 8 = 1.875$$

$$E = 15 / 8 = 1.875$$

$$E = 9 / 8 = 1.125$$

$$\begin{aligned} \chi^2 &= (2 - 3.125)^2 / 3.125 + (2 - 1.875)^2 / 1.875 + \\ &\quad (3 - 1.875)^2 / 1.875 + (0 - 1.125)^2 / 1.125 \\ &= 2.88 \end{aligned}$$

$$\text{Chi-square} = 2.88$$

→ For gender, observed frequencies are:

$$O(\text{gender M, high}) = 2$$

$$O(\text{gender M, low}) = 2$$

$$O(\text{gender F, high}) = 3$$

$$O(\text{gender F, low}) = 1$$

Expected frequencies :

$$E = 2.5$$

$$E = 4 \times 3 / 8 = 1.5$$

$$E = 2.5$$

$$E = 1.5$$

$$\chi^2 = (2 - 2.5)^2 / 2.5 + (2 - 1.5)^2 / 1.5 + (3 - 2.5)^2 / 2.5 + (1 - 1.5)^2 / 1.5$$
$$= 0.53$$

$$\text{Chisquare} = 0.53.$$